Rent or Toxicity?

Strategic exploitability of learning market makers with public wallet information

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6.S974 Games, Learning, and Security — Workshop 1
Example 4: an algorithmic market with adaptive agents

The system: a market where everyone can see who is trading

Hyperliquid, an on-chain exchange

- **Makers** post standing offers to buy and sell (a *quote*); **takers** accept them (a *fill*). The public record of fills is the *tape*.
- Every order and both sides of every fill carry a **public wallet address**.
- Prices live on a coarse grid (one step = a *tick*); same-price orders are served first-in-first-out.
- Takers are **informed** (trade only when the maker will lose) or **uninformed**. Flow with many informed takers is *toxic*.

The puzzle

- Independent learning makers settle on quotes *wider* than the competitive one, without communicating: a **rent** (tacit collusion).
- But a wide quote is also the correct charge for toxic flow.
- On the tape, **rent and toxicity look identical**.

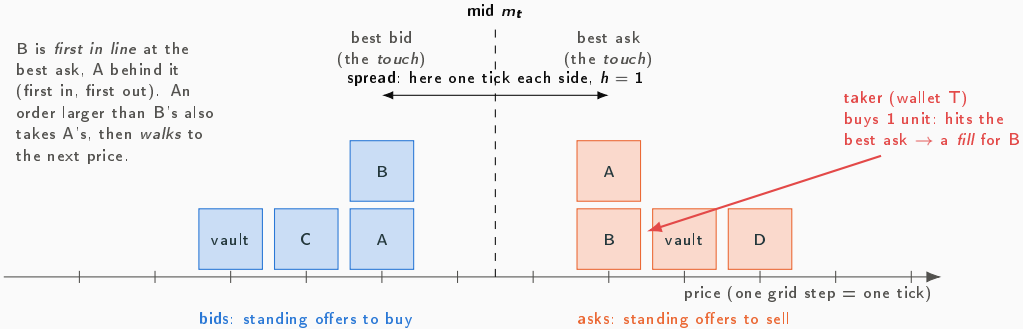
Three players, three decisions

the *incumbents*: how wide; whether to stay in the queue

an *entrant*: which quotes to undercut, which fills to take

the *venue*: public identities? tick size? pay for position?

The book, in one picture



What becomes public: the fill (buy, price, wallet T, wallet B) at once; what the asset is worth afterwards only as the price moves. B's payoff on the fill is price $- V_t$: positive if T was a noise trader, negative if T knew the price was about to rise (*adverse selection*).

On Hyperliquid's major books the spread is one tick essentially always. So a maker cannot earn more by quoting wider; it can only earn by being in line for the *right* fills — which is why the queue, the wallet labels and the identity of T matter.

What we want to learn

RQ1 **What is public wallet identity worth** to an entrant, beyond the anonymous tape — and through which mechanism?

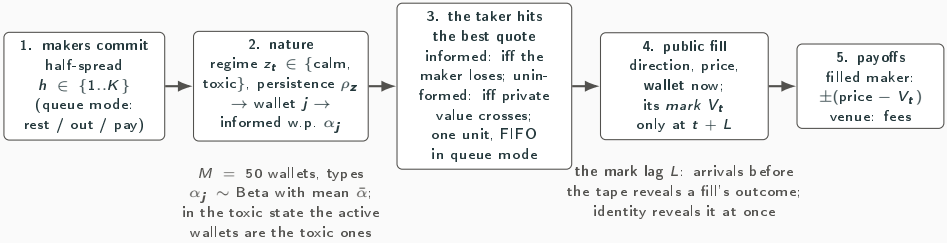
RQ2 **How exploitable is a learned cartel's rent?** How much can an adversary take or destroy, and does more rent mean more vulnerability?

RQ3 **When the grid pins the quote at one tick**, where does rent live, and which venue and maker levers decide who holds the toxic flow?

Why it is a game. The entrant's weapon — undercutting, or taking only the benign fills — is exactly what the makers' learned *reaction function* is tuned to punish; and the venue decides what each side is allowed to know.

Headlines today: the rent is fully exploitable by an entrant that solves the cartel's reaction function; identity's value is a lag story; and on the real book the makers who earn the spread are the ones avoiding the toxic wallets.

One period of the game



Who decides what, with what information

- **Incumbents** ($N = 2, 3$): tabular Q-learners, state = (own last h, best rival's last h); identity-blind by construction. 2×10^4 training episodes, then *frozen*.
- **Entrant**: information set is a treatment, $\mathcal{F}^{\text{anon}}$ (tape) $\subset \mathcal{F}^{\text{id}}$ ($+$ wallets) $\subset \mathcal{F}^{\text{oracle}}$ ($+$ regime); a Bayesian filter gives α_{next}.
- **Venue**: identities public or not; tick; priority fee f.

Horizon and payoffs

- $T = 100$ periods per episode; regime chain and wallets persist across episodes.
- Market-wide expected payoff of a best quote h against informed share α, closed form: $\Pi(h; \alpha) = (1 - \alpha)\, 2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$, $x = h\theta$
- General-sum: the *defenders' loss* is the statistic, not the attacker's gain.

Benchmarks are closed-form, so "rent" is measured, not estimated

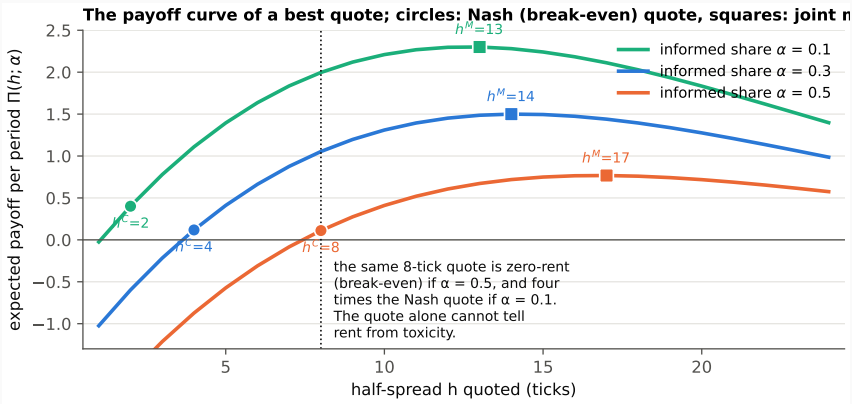
- $h^C = \min\{h : \Pi \geq 0\}$: the **Nash** (competitive, break-even) quote — any wider quote is profitably undercut.
- $h^M = \arg\max_h \Pi$: the **joint-monopoly** quote.
- $\bar{\alpha} = 0.1, 0.3, 0.5$: $(h^C, h^M) = (2, 13), (4, 14), (8, 17)$; the monopoly rent $\Pi^M - \Pi^C = 1.90, 1.38, 0.66$.
- **Rent index** $C = \dfrac{\Pi_{\text{nominal}} - \Pi^C}{\Pi^M - \Pi^C} \in [0, 1]$: where a frozen cartel sits between Nash (0) and monopoly (1).
- With adversary A: **best-response loss** $L^A_{\text{BR}} = \Pi_{\text{nominal}} - J(A)$; **fraction of rent destroyed** $\kappa_A = L_{\text{BR}}/(\Pi_{\text{nominal}} - \Pi^C)$. "Rent units" = divided by $\Pi^M - \Pi^C$.

A worked example ($\bar{\alpha} = 0.3$, tick 0.5)

- Two incumbents at $h = 14$ split $\Pi(14) = 1.50$.
- An entrant at 13 takes every fill: earns $\Pi(13) = 1.49$; incumbents earn 0.
- They answer at 12; the entrant at 11 earns 1.39 . . . the chase ends at $h^C = 4$, $\Pi = 0.12$.
- **A learned cartel is a reaction function that does *not* chase every time.** That restraint is the rent — and the attack surface.

Queue mode (the one-tick book): spread pinned at h; actions rest / out / rest-and-pay; FIFO. All value is in *which fills you take*: at h^C a fill is worth $+1.10$ in the calm state, -2.26 in the toxic one.

Why rent and toxicity look alike: the payoff curve



- Every curve is $\Pi(h;\alpha) = (1-\alpha)\,2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$: uninformed takers pay the half-spread $x = h\theta$, informed ones cost μ when they trade, and trade less often as x grows.
- More toxic flow (larger α) pushes the whole curve down and both benchmarks to the right. An observer who sees only the quote cannot locate it on the right curve — the entrant's first job is to estimate α from the outcomes of its own fills (the *markout* rule), or from *who* is trading.

The prototype, in one command

`python experiments/demo.py` (seconds)

1. prints the benchmarks: $h^C = 4$, $h^M = 14$, $\Pi(h)$
2. plays the shipped two-maker cartel alone: $C = 0.86$
3. enters three adversaries under common random numbers
4. traces the exact best response, period by period

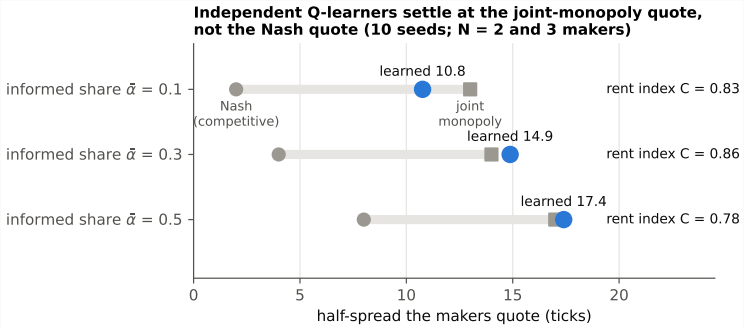
Package (`github.com/pradeepu191/wargame`, 59 tests): `sim/` environment, learners, entrants, queue mode, benchmarks; `experiments/` every result below; `analysis/` the Hyperliquid pipeline (recorder, scoring, identity selection); `results/REPLICATION_NOTES.md` logs each result with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                          cartel loses  entrant earns  kappa_A
  undercut (one tick inside)                 0.41           0.33     0.48
  markout (undercut if rent inferred)        0.47           0.41     0.55
  exact best response                        0.94           1.09     1.10

Trace: exact best response vs the cartel (quotes in ticks)
   t  inc0  inc1  entrant   taker          filled   payoff
   0    15    24       12   noise     buy  entrant   +1.21
   1    23    16       13   noise     buy  entrant   +7.86
   2    20    14       13   no trade
   5    21    19       14   informed  buy  entrant   -5.77
   7    18    23       14   noise     buy  entrant  +11.79
```

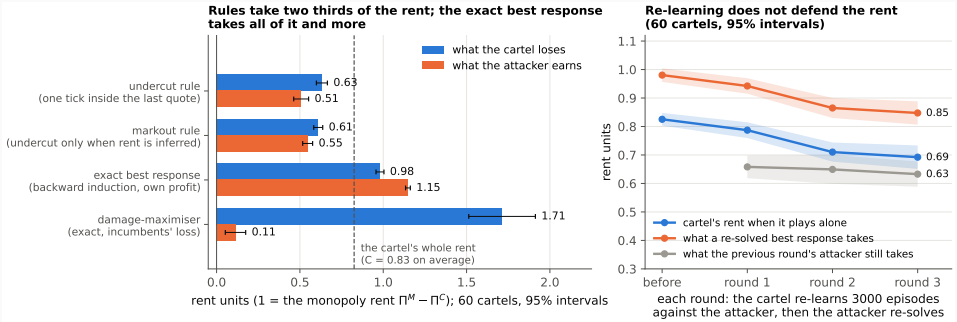
The incumbents answer each entrant quote with their learned reaction; the entrant sits one tick inside what they are *about to* quote and holds them where they never trained.

Evidence 1 — learners coordinate above the Nash point



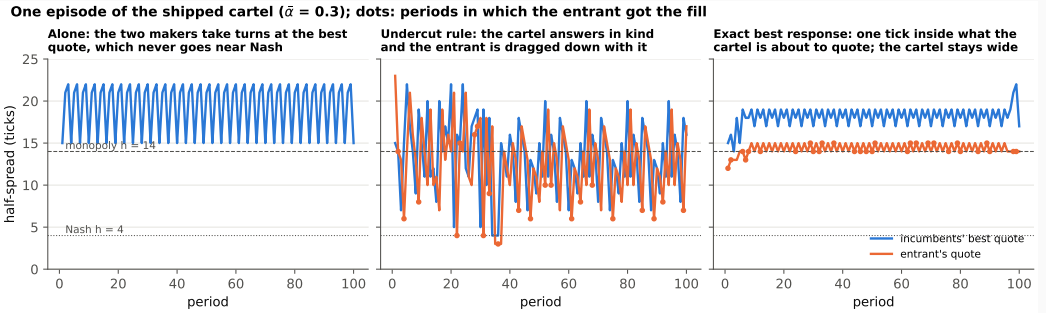
- Replicates Colliard–Foucault–Lovo; $C = 0.71$–0.87 in every cell of a grid over N, learning rate, initialisation.
- Mechanism is **restraint**: in 55–83% of visited states undercutting would pay this period; in 96–100% of those the learner forgoes it. Fills rotate; an undercut is answered in kind and the cartel re-forms.

Evidence 2 — the rent is fully exploitable (RQ2)



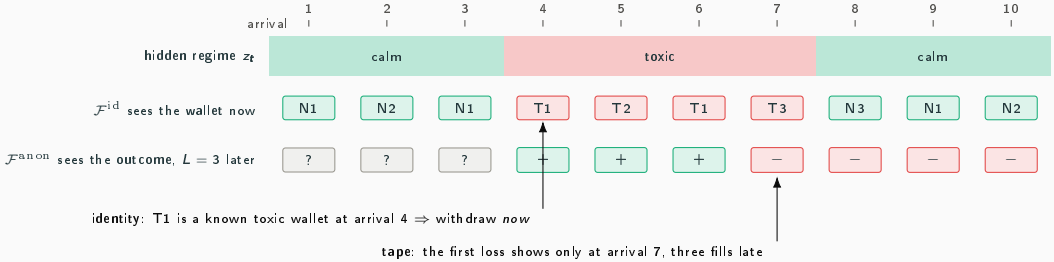
- Rule adversaries destroy a flat $\kappa_A \approx 0.7$ of the rent whatever C (180 cartels). **The exact best response** — backward induction over the frozen cartel's reaction function — takes it all: $L_{\mathrm{BR}} > C$ in 60/60 cartels, the entrant earns the monopoly profit, $\kappa_A = 1.1$–1.3.
- **How:** it undercuts the *reaction*, not the last fill, and holds the cartel in states it never trained on, where its response is to quote *wider*. The damage-maximiser steers it *tight*, inside break-even: 2–3× the rent destroyed at zero cost.

Steering, period by period



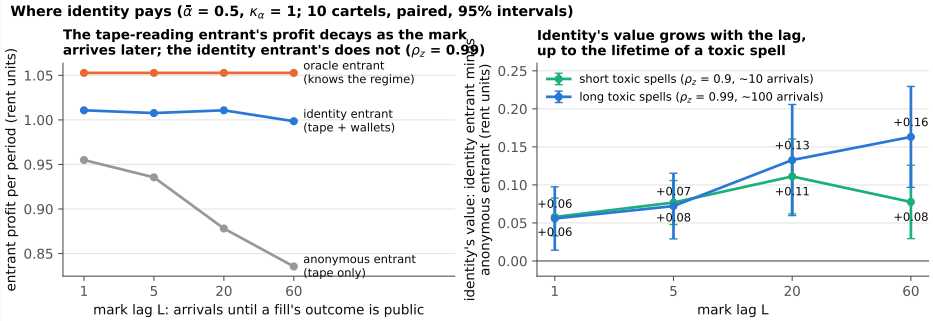
- Left: the learned cartel is a *rotation* — the two makers alternate at the best quote and never go near Nash. Middle: the undercut rule gets a price war: every undercut is answered, both sides slide toward break-even and the entrant's fills (dots) are at ever worse prices.
- Right: the exact best response never provokes the war. It quotes one tick inside the reaction it has *predicted*, so the cartel's learned answer is to stay wide, and the entrant takes nearly every fill at ≈ 14 ticks — the monopoly profit, for itself.

Why identity has value: a spell of toxic flow, seen two ways



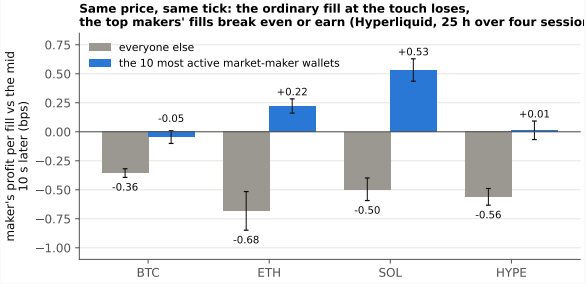
- Informed trading comes in **spells**; the toxic wallets are the ones active in a spell.
- A wallet label identifies the spell at the first fill; the anonymous tape identifies it only when the fills' outcomes become public, L arrivals later.
- So identity is worth $\approx$ (what you lose during the L fills before the tape catches up) — zero if $L = 1$ or if the spell is too short to matter, and growing with L up to the spell's lifetime $\approx 1/(1 - \rho_z)$.
- Hyperliquid's own analogue of L: a fill's outcome is only knowable once the price has moved, seconds to minutes; the wallet is on the fill at once.

Evidence 3 — what identity is worth, and why (RQ1)



- The proposal's mechanism (the same wallet trades again) is worth nothing on the data: same-wallet repeats are 2–5%.
- The mechanism that survives: informed flow comes in **spells**; a wallet's fill reveals the spell *before* the price does. Identity's value is zero where even the toxic state leaves an undercut profitable ($\bar{\alpha} \leq 0.3$), and grows with the mark lag up to the spell's lifetime.

Evidence 4 — Hyperliquid: the price cannot move, and someone still earns the spread

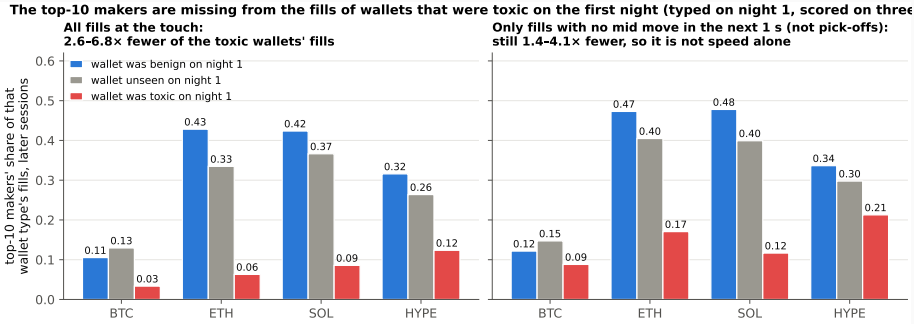


Four best-bid/ask sessions over two weeks (25 h, 490k scored fills) + 449 h of fills with both wallets.

- The quoted spread on BTC/ETH/SOL is **one tick essentially always** (p90 = median). Rent cannot live in the price.
- A fill at the touch pays that tick, then **loses 0.2–0.4 bps** to the mid within 10 s, on every coin. The taker's cost is in depth: one order in ten walks the book and pays twenty ticks.
- The ten most active maker wallets (chosen on activity, scored on outcome) earn **0.3–1.0 bps more per fill at the same price**; their ranking persists across sessions (rank corr. 0.53–0.80).

Rent lives in *which fills you get*, not in the quote.

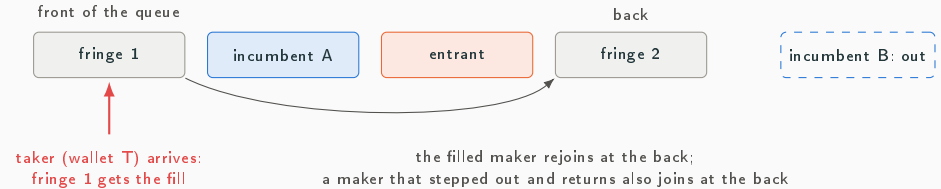
Evidence 5 — how they do it: identity, not the tape



- They never withdraw after a stressful spell at any timescale (30 s–5 min; correlations −0.11 to +0.13). They are absent from the *wallet*, not the *moment*: the gap is the same inside calm and stressed tape.
- Classified on the first night, scored on three later sessions over two weeks (toxic wallets cost 1.1–2.1 bps more there: the typing holds). **RQ1's mechanism, observed on the real book.**

The queue game, in one picture (the one-tick book)

Period t: the spread is pinned at one tick; each maker chooses rest or out; those resting stand in line, first in first out. The taker takes the front.

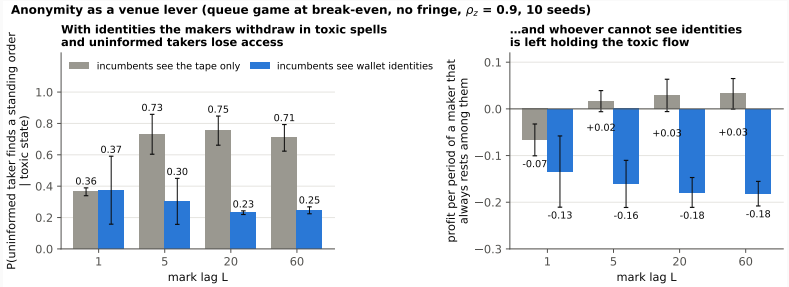


Payoff of a fill at break-even h^C against a taker of type α: $(1-q)\,x - q\,\mu$, with q the probability the fill was informed. On average $+0.15$; in a calm spell $+1.10$; in a toxic spell -2.26. All the value is in *which fills you take*, and time priority rations them.

What a selector does. Incumbent B stepped out because its forecast says the next taker is toxic; it gives up its place in line to avoid the fill. The identity entrant does this on the wallet label; the tape-reading entrant on outcomes L late.

Who pays. Whoever rests through the toxic spell — the *fringe* (retail orders, the vault) — takes those fills. The priority fee (rest-and-pay) lets a maker buy the front of the line: an all-pay auction for the calm fills.

Evidence 6 — the levers decide who holds the toxic flow, not whether the rent survives (RQ3)



- **Anonymity** is a transfer from makers to uninformed takers in stress; **identity** the reverse. Neither defends the rent.
- **Priority fee**: an all-pay auction for position; at the venue's optimum it collects 2–4× the makers' rent and leaves every maker at or below zero; the identity-blind fringe pays.
- **Makers' own levers**: only the identity-aware floor improves both profit and resilience, only where the toxic regime bites, and it still leaves the adversary 75–80% of its take.

Plan for Workshop 2

Attacks and learning

1. **The attacker with public information only**: identify each incumbent's reaction function from the tape (every answer to (own, rival) is public; the entrant sets "rival" with its own quote), then solve. Report its gap to the exact best response vs the probing budget.
2. **Both sides learning**: PSRO with re-learning cartels and learned entrants — the equilibrium's rent and κ_A; are adversarially trained cartels less exploitable, or just less collusive?
3. **Queue-mode attacks**: steering no-regret incumbents; draining the request budget.

Resilient designs, and data

- Venue levers (anonymity, tick, fee, request allowance) and maker levers on one frontier: (profit alone, profit in the worst 5% of episodes under attack) against all three attackers, plus uninformed takers' access in stress — so "robust" cannot mean "withdrew".
- Weeks of recording (identity-selection gap by coin, hour, regime); the Oct 2025 liquidation cascade; the Dec 2025 order-level archive: *which action* — cancel, queue position, depth — carries the selection.
- Throughout: common random numbers, paired intervals, worst-5% outcomes, train-at-$\bar{\alpha}$/test-at-$\bar{\alpha}'$, held-out adversaries.

What we would value your view on: is identifying a frozen reaction function the right "realistic attacker", or should the cartel be allowed to adapt continuously?

Backup — Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	−0.41 / −0.80	−0.56 / −2.26	−0.27 / +0.98	−0.45 / −0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	−0.64	−1.01	−0.59	−0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	−0.05 / −0.36	+0.22 / −0.68	+0.53 / −0.50	+0.01 / −0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
... on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs.
Wallet types from the first night, scored on the three later sessions.

Backup — the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls $1.2 \to 0.5$ (the undercut-rule level); the re-solved one takes $\geq 90\%$ of the remaining rent in every cartel every round; the cartel's own rent erodes $0.83 \to 0.79 \to 0.71 \to 0.69$.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4–7 ticks, inside break-even ($h^C = 8$–13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7–2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$–0.86, $\kappa_A = 0.65$–0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75–0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup — the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10 — identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93–94% of calm, 8–9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2–5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.